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Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

A Bin Level You Can't Trust: Radar and Ultrasonic in Silos and Hoppers

Field notes from Jonathan Gilmour, independent controls engineer, Bay Area and Northern California.

A silo level that reads full when the loader shows it half, or drops to zero the moment the pack blower kicks on, will burn your night either way. Operators stop trusting the number, so they climb the silo with a tape or a plumb bob, or they run the discharge dry and cook a bearing. Most of the time the material didn't move the way the number says. The sensor got fooled, the mounting is wrong, or the controls are averaging garbage into something that looks smooth and believable. This guide is about telling those apart on the equipment you actually have, at 2am, without pulling the sensor first thing.

Two terms up front so the rest reads clean. **FMCW (Frequency-Modulated Continuous Wave)** is the radar method most modern non-contact level transmitters use: the antenna sweeps a band of frequency and measures the surface distance from the beat frequency of the returned sweep. It handles dust and vapor far better than the old pulse-time radars. **GWR (Guided-Wave Radar)** runs the microwave pulse down a physical probe — a rod or cable hanging into the material — so the signal is guided instead of sprayed into open air; it shrugs off dust and low dielectric better, at the cost of a probe you have to keep clean and clear of buildup. I'll define **deadband / blocking distance** in the near-full section where it bites.

Before anything with a lid on it: this bin can kill you

A silo, hopper, bin, or surge bin is a **permit-required confined space with an engulfment hazard.** People die in these every year, and they die fast. The rules below are not optional and they come before any troubleshooting step that puts you on top of, inside, or opening the vessel.

- **Never enter under a live draw.** If discharge can run — screw, rotary valve, gravity gate, pneumatic pack — flowing material pulls a person under in seconds. There is no swimming out.
- **Never trust the top of a bridged or ratholed load.** Material that hangs up forms a crust (a **bridge**) over an open void, or a vertical chimney (a **rathole**) with caked walls. It looks like solid floor. It collapses without warning when you step on it or when the discharge finally pulls it loose. Assume any hung-up bin is booby-trapped.

- **Lock out fill AND discharge.** Zero-energy state means both directions: the fill conveyor/blower and every discharge device, plus any live-bottom, agitator, bin activator, or air-pad blower. Apply LOTO (Lockout/Tagout), verify the isolation, and try-to-start.
- **Test the atmosphere before entry, from outside, top to bottom.** Silos go oxygen-deficient from material off-gassing, decomposition, or inerting/nitrogen blanketing. Grain, wood, biomass, and organics produce their own hazards. Carry a **bump-tested personal 4-gas monitor** (O2, LEL, CO, H2S) and keep it on you inside, not clipped at the manway.
- **Dust is an explosion hazard.** Combustible dust (grain, sugar, wood, coal, some plastics and metals) suspends and ignites. Mind ignition sources, bonding/grounding, and whether the space is a classified area before you bring a tool or a light in.
- Full confined-space program: written permit, trained attendant outside, tested air, rescue plan, retrieval harness. If your gut says "I'll just pop the hatch and look," that's exactly the move that gets a person hurt. Read the number from the outside first.

Almost everything below can be done from the transmitter head, a laptop, and the ground. Keep it that way as long as you can.

10-minute triage

Fast, safest-first, before you commit to a theory. No entry, no climbing.

1. **Pull the live echo/waveform, not just the 4-20 mA.** Nearly every modern radar or ultrasonic transmitter shows a raw echo profile at the display or in the config software. One look tells you whether the device is tracking a real surface, locked on a false echo, or lost. This is the single highest-value 60 seconds you'll spend.
2. **Compare against a second source right now.** Belt-scale or load-cell totalizer, batch count, last known load-in ticket, or a hand tape reading from an existing gauge port. Establish the error size and sign: reads high or low, by how much.
3. **Watch it through one process event.** Have the dust collector / pack blower cycle, or watch a discharge pulse. If the level jumps or drops in lockstep with a fan, valve, or agitator, you've found an interaction, not a level change.
4. **Check the trend, not the instant.** A sawtooth that snaps up on every fill and creeps down smoothly is healthy. A reading that's flat-lined for hours through known draw is stuck (buildup or a latched false echo). A number that thrashes ±10% with no process reason is noise or a moving false target.
5. **Signal strength / echo confidence value.** Falling amplitude points to buildup on the horn/lens or dust attenuation. A strong echo at the wrong distance points to a structural false target or ringing in the nozzle.
6. **Note whether it fails only near full or near empty.** That's a deadband/blocking-distance or bottom-of-cone problem, covered below, and it's a config fix, not a sensor swap.

Log every one of these with a timestamp. The pattern across a few events is what solves it; a single spot reading rarely does.

Symptom → likely cause → check (and what to log)

Symptom	Likely cause	Check (and log)
Reading stuck flat while bin is known to be drawing down	Buildup/coating on the horn or lens (ultrasonic) or antenna (radar); or transmitter latched on a fixed false echo	Pull echo profile: is the tracked echo pinned at one distance? Log signal amplitude and the frozen distance. Compare to discharge totalizer over the same window.
Level reads lower with dust collector / pneumatic fill running, recovers when it stops	Dust-cloud attenuation of the sound/microwave path (hits ultrasonic far harder than radar)	Trend level vs. blower run signal. Log dB/amplitude drop during the event. If ultrasonic drops out and radar wouldn't, that's your tell.
Level pins at a fixed value near the top no matter how full	Near-full inside the deadband/blocking distance, or antenna seeing the fill stream/nozzle	Log the exact "stuck" distance; compare to the configured blocking distance and antenna face location.
Reads a solid partial level when the bin is actually empty	False echo off cone bottom weldment, discharge cone, agitator, or a fixed structural target below true empty	Empty the bin verified by discharge; capture echo profile of the empty vessel. Log the false-echo distance and amplitude for masking.
Jumpy, thrashing reading with no real level change	False echoes off agitator/bin activator, sidewall ladder/braces, or fill stream crossing the beam	Correlate thrash timing with agitator rotation or fill pulses. Log min/max swing and whether it tracks a moving part.
Sensor and belt-scale/load-cell disagree by a steady offset	Angle of repose: point measurement under a fill or draw cone vs. true averaged volume	Log level at both a just-filled and a just-drawn state. The point reading leads/lags volume by the cone; quantify the gap.
Number believable but always a little off after a fill	Uneven surface / sidewall fill piling; beam sees the peak or the valley, not the mean	Note fill-point location vs. sensor aim. Log reading right after fill vs. after material settles.
Intermittent high-amplitude ghost at a repeatable distance	Nozzle/standpipe ringing (multiple internal reflections in a long or narrow mounting nozzle)	Log the ghost distance; check if it equals the nozzle length or a multiple. Reflectors/ring at the nozzle base is the fix.

Symptom	Likely cause	Check (and log)
Radar reads through to a lower false surface intermittently	Low-dielectric material (light fluffy powder, plastic pellets) giving weak return; radar penetrating to bin bottom	Log dielectric of product and echo amplitude. Candidate for GWR or a lower-frequency/higher-sensitivity radar.
Slow drift high over weeks, cleans up after a wash-out	Gradual coating accumulation on antenna/lens, or condensation/foam on the surface	Log amplitude trend over time; note humidity/temperature and any steam or hygroscopic product.

Reading the failure modes

Buildup and coating. Sticky, hygroscopic, or fine product cakes on an ultrasonic transducer face or a radar horn/lens. On ultrasonic it kills you two ways: the coating damps the transmit pulse and it deadens the ring-down, so the device either loses the echo or reads the coating itself as a near surface. Radar tolerates a thin film but a heavy dielectric cake in front of the antenna throws a strong permanent near-echo. The tell is a reading that stops responding to real draw and a signal amplitude that's been sliding for days. If your product coats — lime, flour, resin, wet fines, anything that clumps — this is your first suspect and the strongest argument for GWR, or better, a non-contact radar with an air-purge connection on the antenna.

Dust-cloud attenuation. A dense dust cloud during pneumatic fill or when the dust collector pulses scatters and absorbs the signal on its way to the surface and back. Sound gets hammered — ultrasonic can drop out entirely in heavy loading because the acoustic energy is absorbed by the particulate and the temperature/density of the cloud shifts the speed of sound your device assumed. FMCW radar walks through the same cloud with far less loss because microwave energy barely notices dust. If your level "goes empty" every time the fill blower or the collector runs and comes back after, you're not watching material, you're watching the cloud. This is often the deciding reason plants move a dusty silo from ultrasonic to radar.

False echoes off structure, agitators, and the fill stream. The beam is a cone, not a laser. Anything it clips — a ladder rung, a weld seam, a sidewall brace, a bin-activator cone, a rotating agitator paddle, or the falling fill stream itself — throws a return. A fixed structural echo gives a believable steady false level; a rotating agitator gives a reading that thrashes in time with the paddle; the fill stream gives a false "spike to full" during load-in. The cure is aim and masking: point the beam at clear surface away from structure and the fill point, then teach the transmitter to ignore known fixed echoes (false-echo mapping / TVT, done with the bin at a known state). Never mask so aggressively that you also blind the device to a real surface at that distance.

Uneven surface and angle of repose. This one is physics, and it's the most common "the sensor is wrong" that isn't. Solids don't self-level. Fill under the inlet builds a cone (angle of repose); draw over the outlet pulls a funnel. A point-measuring beam reads whatever height sits under it — the peak of the fill cone or the valley of the draw funnel — which can differ from the true average volume by several feet in a big silo. If your radar and your belt-scale totalizer disagree by a consistent amount that flips sign between filling and drawing, that's angle of repose, not instrument error. You manage it with sensor placement (mount off-center and aim at a representative point on the pile slope — typically about a third of the radius in from the wall, clear of both the incoming fill stream and the sidewall — rather than at the center peak the fill cone throws up or the funnel the draw pulls down), by trending rather than trusting the instant, and by reconciling to a weight-based source for true inventory.

Foam and condensation. Less common in dry bins, real in some processes — a hygroscopic powder pulling moisture, steam off a warm product, or foam on a slurry surge tank feeding a bin. Condensation on a radar lens acts like light coating; foam absorbs and scatters both sound and low-power microwave. If the problem tracks humidity or temperature and clears when things dry out, look here before you blame the electronics.

Near-full and near-empty: deadband, blocking distance, and the cone. Deadband (also called blocking distance or near-zone) is the region right in front of the antenna or transducer that the device is told to ignore — it exists because the sensor can't cleanly resolve echoes too close to its own face, and because you don't want it locking onto the nozzle or the fill stream. If material rises into that blocking distance, the reading pins at the top of the measurable range and won't budge until level drops back out — so a silo can read "95% and holding" while it's actually packed to the roof. Mount high enough, and set blocking distance no larger than you truly need. At the bottom end, the last portion of a cone often can't be measured because the beam hits the discharge weldment or the residual heel; know your true empty and don't scale the 4 mA to a level the sensor can't actually see.

Nozzle and standpipe ringing. Mounting the sensor on a long or narrow nozzle (a standpipe) turns the nozzle into a resonator: the pulse bounces off the nozzle walls and lip, throwing ghost echoes at distances that match the nozzle length. You'll see a repeatable false target that doesn't move with level. Fixes are a shorter, wider nozzle, a nozzle cut at an angle at the bottom, a beam-focusing antenna extension that clears the nozzle, or in bad cases relocating the mount. If the ghost distance equals your nozzle length, stop looking at the material.

When to change technique

You've aimed, masked, purged, and it still lies. Now it's worth changing the measurement, not just retuning it.

- **Heavy buildup or low-dielectric product → guided-wave radar (GWR).** Guiding the pulse down a probe recovers weak returns from fluffy, low-dielectric powders and rides through dust. The trade is a probe in the material: abrasion, pull-down force in a deep silo, and its own coating to manage. Good for lime, cement, plastic pellets, grain in the right build. Not for very tall silos where cable tension and buildup on the probe become their own headache.
- **Dust and vapor with a clean shot → non-contact FMCW radar.** For a dusty silo that eats ultrasonic, an 80 GHz FMCW transmitter with a narrow beam is usually the answer: it sees through

the cloud, the tight beam dodges structure, and there's nothing hanging in the product. Add an air purge on the antenna if the product coats.

- **Any critical high/low → add an independent point-level switch.** This is the one I push hardest. A vibrating-fork, rotary-paddle, or capacitance point switch at the high-high and low-low positions gives you a physical, independent cross-check that does not share a failure mode with the continuous transmitter. When the radar and the point switch disagree, you know something's wrong before you overfill or run dry. For a plugged-chute or overfill interlock, the point switch should be the interlock, with the continuous level as information.

The controls angle

The instrument is only half of it. A clean sensor feeding a sloppy control scheme still gives you a level you can't trust.

Scaling and damping. Scale to the vessel's true measurable range — real empty (accounting for the cone/heel the sensor can't see) to real full (below the blocking distance) — not to a nameplate volume. Then damp with intent. A little filtering smooths the sawtooth of normal fill/draw; too much and you've hidden a real fast event, like a bridge collapse dumping level, behind a long time constant. Set damping to the slowest real process rate you need to see, no slower. And know that averaging point measurements never gives you true volume when the surface is coned — damping smooths the number, it doesn't fix the physics.

Keep interlocks independent. High-level and low-level protective interlocks must not derive from the same signal, sensor, or scaling as the number the operator watches for control. If the continuous radar drives both the operator display and the overfill trip, one coated antenna takes out your protection and your indication together. Independent point switches on separate wiring, separate inputs, separate logic. This is basic separation of the control layer from the protection layer, and it's exactly where a lot of retrofits quietly cut a corner.

Dust-collector and fill-cycle interactions. If level readings move with the collector or the fill blower, either the process is genuinely disturbing the surface/air path or you're reading the cloud. Where it's a measurement artifact, the clean fix is to gate or hold the reading during the disturbance — freeze the last good value while the blower runs, resume after it settles — rather than letting a filter smear the dropout into a wrong number. Coordinate that hold with the actual fill-cycle logic so you're not holding through a real level change.

Reconcile to weight. For inventory that matters — reconciliation, custody, production accounting — trust a belt-scale or load-cell totalizer over a point level in a coned bin, and use the continuous level for rate and for high/low action. Cross-check them daily: a growing divergence between summed weight and indicated level is an early warning that the sensor is coating up or drifting, often before an operator would notice. Log the divergence; the trend is the diagnostic.

Closing note

Most untrustworthy bin levels come down to one of a handful of things — a coated horn, a dust cloud, a false echo off structure, or a coned surface the point sensor was never going to average — and you can sort them from the ground with the echo profile, a trend, and one independent cross-check. Change the technique only after you've proven aim, masking, and purge won't hold. Where an independent second look tends to earn its keep is a level that's been "recalibrated" by three different vendors and still lies — that's usually a mounting or separation-of-interlock problem nobody owned, not a bad transmitter. And whatever you do, read it from outside the vessel: the bin does not care that you were only going to look for a second.